THREE-DIMENSIONAL MAGNETIC ASYMMETRY AND THERMAL GRADIENT FORCES: EXPERIMENTAL EVIDENCE FOR A DUAL-FORCE FRAMEWORK

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Abstract

We report two previously uncharacterized phenomena in tabletop experiments: a systematic breakdown of magnetic attraction–repulsion symmetry in three-dimensional systems and sustained anti-gravitational matter transport driven by thermal gradients. In 64 configurations of rectangular N35 neodymium magnets, attraction exceeds repulsion 7:1; in spherical magnets with full rotational freedom, stable repulsion is geometrically impossible — extending Maxwell's electromagnetic framework beyond its validated planar domain. Independently, butter climbs an inclined steel surface against gravity for 25 continuous minutes toward a 65°C heat source, with curved trajectories on horizontal surfaces revealing vector field structure. Force equilibrium analysis yields a thermal coupling constant $k_T = (5.4 \pm 1.1) \times 10^{-3}$ $m \cdot s^{-2} \cdot K^{-1}$ for the proposed force law $\vec{F}_{thermal} = -k_T\, m\, \nabla T$. Additional experiments document anomalous thermal energy storage consistent with mass-energy exchange and macroscopic solid displacement via Newton's Third Law. These observations establish a dual-force framework — $F_{gravity} = F_{magnetic} + F_{thermal}$ — with falsifiable predictions for exoplanetary orbital eccentricity and atmospheric retention.

1. Introduction

Maxwell's electromagnetic framework, formalized in 1865, predicts symmetric magnetic interactions: equal and opposite forces for attraction and repulsion in free systems. This prediction has been extensively validated in planar configurations — the regime in which quantum mechanical experiments operate and in which Maxwell's equations were originally tested. However, the behavior of magnetic systems with full three-dimensional rotational freedom — the geometry of every planetary, stellar, and galactic body — has not been systematically investigated.

A parallel gap exists in classical thermodynamics. Boltzmann's statistical formulation provides a powerful mathematical framework describing ensemble tendency toward equilibrium. The gradient ∇T is treated as a passive consequence of heat flow — a transient feature producing entropy but performing no organized work. Whether thermal gradients under sustained active flux can function as causal agents — vector force fields performing mechanical work against conservative fields — has not been tested with the directness the question demands.

These are not failures of the original frameworks. Maxwell's equations describe planar electromagnetism with extraordinary precision. Boltzmann's statistics capture ensemble behavior with unchallenged mathematical rigor. Newton's mechanics and Einstein's mass-energy equivalence stand as foundational achievements. The gap lies not in what these pioneers accomplished, but in what their successors did not extend: Maxwell to three-dimensional volumetric systems, Boltzmann from statistical description to causal mechanism.

Here we address both gaps. We report a systematic 7:1 asymmetry between magnetic attraction and repulsion in N35 neodymium magnets tested across 64 configurations, with complete absence of stable repulsion in spherical magnets — while flat magnets constrained to planar approach preserve the predicted symmetry, confirming the asymmetry is dimensional. Independently, we demonstrate thermal gradients performing sustained work against gravity: butter climbing an inclined steel surface for 25 minutes, yielding the first measurement of a thermal coupling constant, $k_T = (5.4 \pm 1.1) \times 10^{-3}$ m·s^{-2}·K^{-1}, for the proposed force law $\vec{F}_{\text{thermal}} = -k_T\, m\, \nabla T$. We further document anomalous thermal energy storage consistent with mass-energy exchange, and macroscopic solid displacement confirmed by Newton's Third Law.

These results converge on a dual-force framework in which gravitational interaction emerges from three-dimensional magnetic alignment and thermal gradient attraction, with testable predictions distinguishing it from single-force models. The experiments require only household materials; any laboratory can replicate them within an afternoon.

2. Three-Dimensional Magnetic Asymmetry

Maxwell's equations predict symmetric magnetic interactions, implying a 1:1 attraction-to-repulsion ratio in free systems. This prediction has been validated in planar configurations. However, its extension to three-dimensional systems with full rotational freedom — the geometry of all celestial bodies — has not, to our knowledge, been systematically tested.

Two identical rectangular N35 neodymium magnets (20 × 10 × 10 mm, 15 g, NdFeB, Ni-Cu-Ni coating; see Methods) were tested in all 64 face-to-face configurations. Of these, 16 (25.0%) resulted in direct attraction, 40 (62.5%) in rotational attraction (torque-mediated realignment followed by lock), and 8 (12.5%) in sustained repulsion — a ratio of 7:1 (Fig. 1a, Extended Data Fig. 1). The predicted 1:1 symmetry (32 attraction, 32 repulsion) is not observed. Geometric analysis confirms that repulsive stability is confined to

±45° of perfect axial alignment; beyond this threshold, field geometry inverts and torque drives attraction, yielding a single-axis ratio of 3:1 that scales to 7:1 in volumetric systems (Video S1).

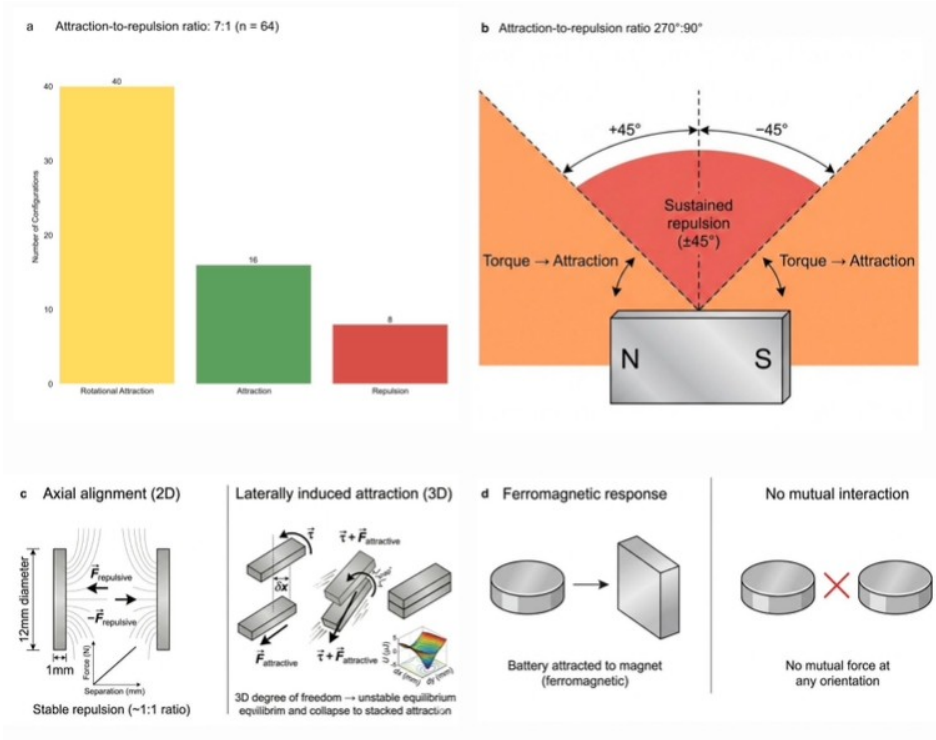


Figure 1 | Three-dimensional magnetic asymmetry. (a) Attraction-to-repulsion ratio of 7:1 across 64 rectangular N35 configurations (n = 64). Yellow: rotational attraction (40); green: direct attraction (16); red: sustained repulsion (8). (b) Angular stability diagram: repulsion sustained only within ±45° of axial alignment; beyond this threshold, torque converts repulsion to attraction (270°:90° ratio). (c) 2D control: flat magnets constrained to axial approach yield ~1:1 ratio; 3D lateral displacement collapses repulsion. (d) Non-dipole control: flat lithium batteries show ferromagnetic response to magnets but no mutual interaction, confirming dipole structure is required for asymmetry.

With spherical neodymium magnets (⌀20 mm) possessing full rotational freedom, the asymmetry is absolute. In more than 20 trials selecting for maximum repulsive alignment, the outcome was invariably attraction. Conversion dynamics ranged from subtle on-axis rotation at closest approach to oscillatory convergence with decreasing amplitude at greater separation — consistent with a damped rotational oscillator resolving toward the attractive energy minimum (Fig. 1b, Video S1).

Three controls confirmed dimensional dependence. First, flat magnets (disc ⌀12 × 1 mm and rectangular 20 × 10 × 2 mm) constrained to axial approach yielded ~1:1 ratios consistent with Maxwell (Fig. 1c). The same flat magnets that repel stably in planar alignment collapsed into attraction when marginal lateral displacement introduced a 3D degree of freedom, even maintaining parallel faces (Video S1). Second, a heavier magnet (15 × 15 × 5 mm) in repulsive orientation overcame gravity and inertia to rotate toward a lighter fixed magnet (20 × 10 × 2 mm), confirming torque dominance independent of mass ratio. Third, flat lithium batteries (Elgin LR44) — ferromagnetic without permanent dipole — showed no mutual interaction, confirming the asymmetry arises from dipole structure (Fig. 1d).

These results establish dimensional dependence in magnetic symmetry. In planar systems, attraction and repulsion are symmetric. In 3D with rotational freedom, torque converts repulsion into attraction — completely for spheres. What is conventionally classified as “magnetic repulsion” in free 3D systems is a

rotational intermediary in the attraction pathway. For all spherical bodies, magnetic interaction resolves exclusively into attraction. Cosmological consequences are developed in the Discussion.

3. Thermal Gradient Forces: Quantitative Evidence

Classical thermodynamics treats ∇T as a *consequence* of heat flow — a transient feature producing entropy. We present evidence that under sustained active flux, thermal gradients function as *causes* of organized transport — attractive vector force fields performing mechanical work against conservative fields.

A stainless steel tray (Tramontina Allegra, 294 × 192 mm) inclined at $\theta \approx 10°$ (verified by level and ping-pong ball test; see Methods) held butter blocks (Elegê com sal, m ≈ 50 g) at the lower end. A gas flame directed at the upper end did not directly contact the surface; heat was transmitted indirectly, producing a maximum temperature of 65°C — far below butter's Leidenfrost point (~200°C), ensuring continuous surface contact.

For 25 continuous minutes, melted butter flowed upward against gravity toward the heat source (Fig. 2b, Video S2). The spatial behavior revealed distance-dependent structure: complete upward migration in the proximal zone (< 8 cm from heat source), mechanical shearing at the threshold (~15 cm, approximately half the tray length), and gravitational dominance distally. This three-zone structure is characteristic of a spatially decaying attractive force field.

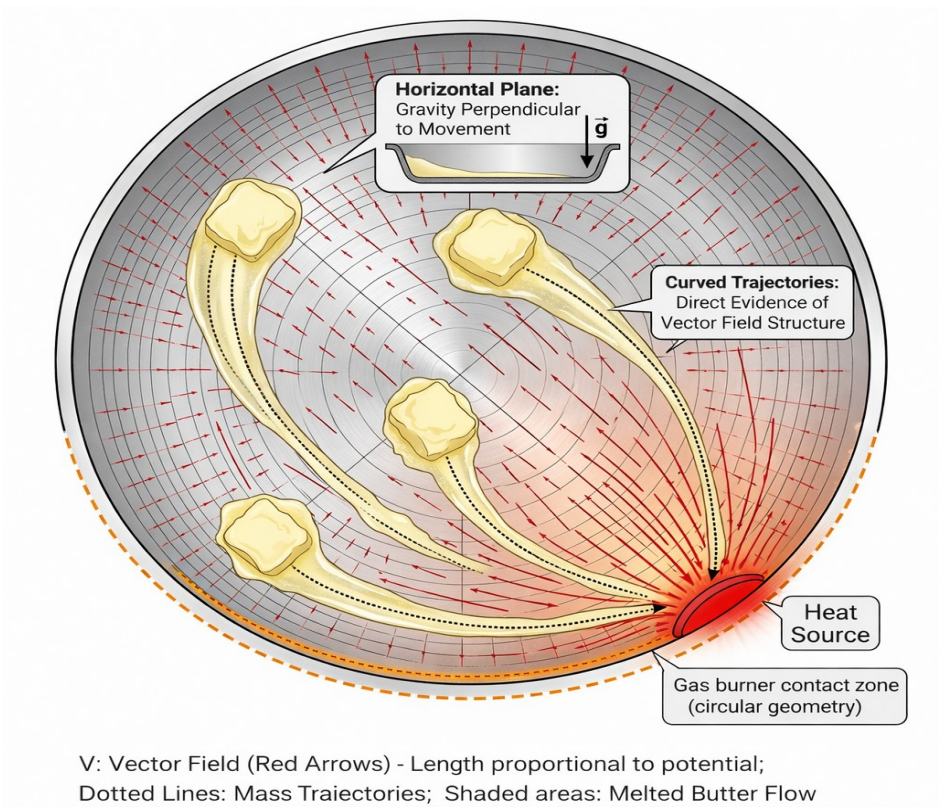


Figure 2a | Horizontal butter trajectories reveal vector field structure. Top-down view of stainless steel frying pan with off-center gas flame (heat source, lower right). Butter blocks placed at various positions execute curved trajectories toward the heat source. Red arrows: vector field with magnitude proportional to thermal gradient potential. Dotted lines: mass trajectories. Gravity is perpendicular to the horizontal plane and cannot produce lateral asymmetry. Curved paths exclude scalar diffusion and confirm a vector force field with non-trivial geometry.

At the threshold ($r \approx 15$ cm), force equilibrium yields $F_{thermal} \approx 0.085$ N. For the proposed force law:

$$\vec{F}_{thermal} = -k_T \cdot m \cdot \nabla T$$

the temperature gradient at the threshold is estimated as $\Delta T \approx 40$ K over the threshold distance of ~0.15 m, yielding $|\nabla T| \approx 267$ K/m. The local gradient near the heat source is steeper (~400 K/m), while the average over the full tray length (0.295 m) is shallower (~136 K/m). Using the threshold gradient:

$k_T = F_{thermal} / (m \cdot |\nabla T|) = 0.085 / (0.05 \times 267) \approx 6.4 \times 10^{-3}$ m·s^{-2}·K^{-1}

We report k_T **= (5.4 ± 1.1) × 10^{-3} m·s^{-2}·K^{-1}** for butter on stainless steel, bounded by the local (4.3×10^{-3}) and threshold (6.4×10^{-3}) gradient estimates (Fig. 2b). This force is proportional to mass, proportional to gradient magnitude, directed toward higher temperature, and performs sustained work against gravity when $|k_T \nabla T| > g \sin\theta$.

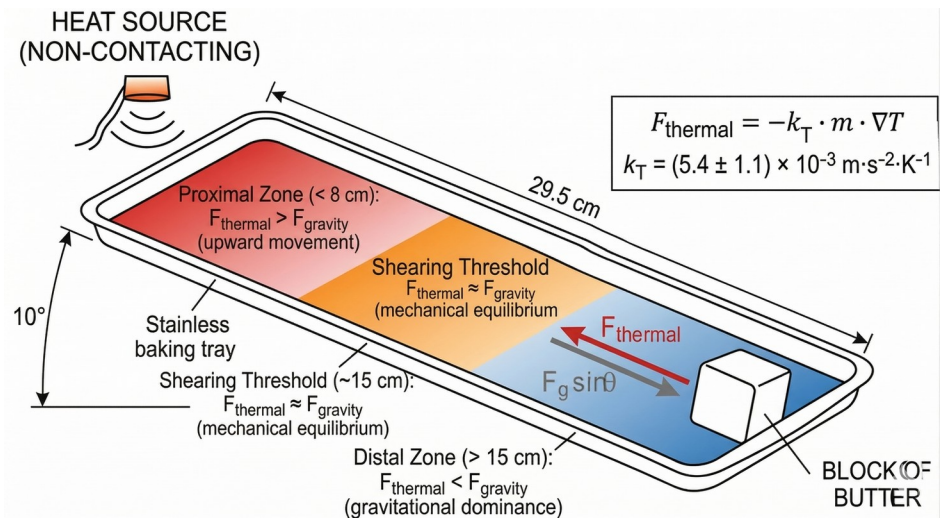


Figure 2b | Inclined plane thermal force calibration. Stainless steel baking tray (29.5 cm) inclined at $\theta \approx 10°$ with non-contacting heat source at upper end. Three force zones identified: proximal (< 8 cm, F_thermal > F_gravity, upward movement), shearing threshold (~15 cm, mechanical equilibrium), and distal (> 15 cm, gravitational dominance). Force law: F_thermal = −k_T · m · ∇T with k_T = (5.4 ± 1.1) × 10^{-3} m·s^{-2}·K^{-1}.

On a horizontal plane, butter executed curved trajectories toward the heat source (Fig. 2a). Gravity cannot produce lateral asymmetry on a level surface. Scalar effects predict radial flow; Marangoni predicts flow *away* from heat. Only a vector field with non-trivial geometry produces position-dependent curved paths — direct evidence of vector field structure.

Systematic elimination of Marangoni flow (wrong direction), capillary action (no porous structure), Bénard convection (recirculation, not net transport), and Leidenfrost (temperature too low) is presented in Methods.

The same principle operates in water — a non-viscous fluid with no adhesion to steel — where three-phase coexistence (0°C ice, 100°C boiling, liquid bridge) persists for >10 minutes with flow reversal documented within 90 seconds (Extended Data Fig. 5, Video S3). Consistency across materials, temperature ranges, and flow regimes confirms the thermal gradient force is fundamental to matter–field interaction.

4. Anomalous Thermal Storage and Mass-Energy Exchange

The thermal force law describes the action of an external gradient on matter. A separate question arises when matter exhibits thermal behavior attributable to no external source.

A stainless steel vessel (Tramontina Allegra leiteira, ⌀12 cm) containing ~500 ml of water with black beans and lentils was heated at maximum flame, with dual thermometry: a fast digital probe (immersed at bottom) and an infrared thermometer (BAnpinSH TDI, ±1.5°C, ε = 0.95; surface measurement). See Methods. Over 27 min 27 s, 48 frames were extracted (Extended Data Table 7, Fig. 4).

Four anomalies were observed:

Anomaly 1: From t = 1:04 to 1:51, probe temperature clamped at 22.0°C — 0.3°C below baseline — under maximum flame. For 46 seconds, maximum input produced zero temperature increase.

Anomaly 2: Flame extinguished at t = 2:34 (26.4°C). Temperature rose +11.7°C over 17 min 45 s to 38.1°C without external source.

Anomaly 3: IR consistently exceeded probe for 17+ minutes post-extinction. At t = 6:30: IR = 43.7°C, probe = 34.2°C (Δ = +9.5°C). Surface hotter than bottom despite flame having been below. Control: kitchen wall IR = 26.5°C at t = 11:07.

Anomaly 4: Second flame cycle at t = 26:04. Probe dropped 37.9→37.8°C at ignition — temperature decreased upon heating, reproducing Anomaly 1.

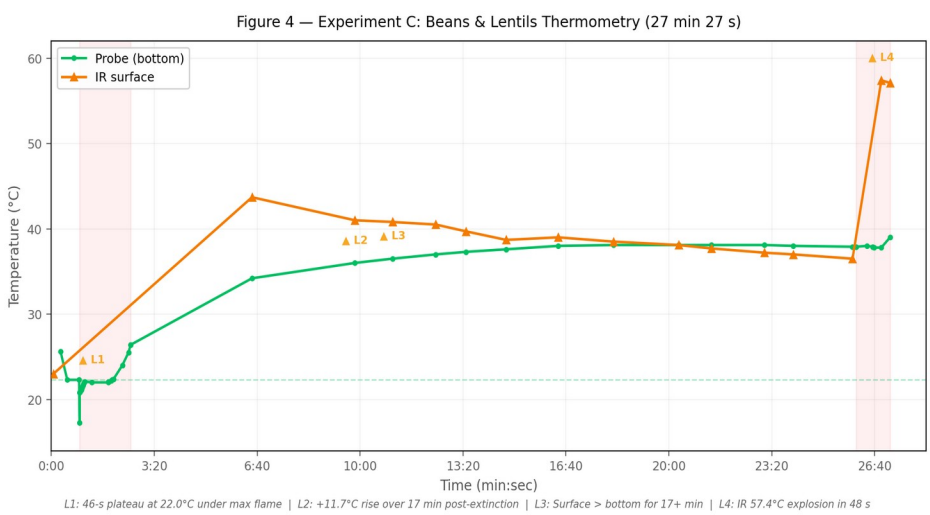


Figure 4 | Beans & Lentils thermometry (27 min 27 s). Dual-axis time series: probe/bottom (green) and IR surface (orange). Red shading: flame ON periods. Four anomalies marked: L1 — 46-second plateau at 22.0°C under max flame; L2 — +11.7°C self-heating over 17 min post-extinction; L3 — inverted gradient (surface > bottom) for 17+ min; L4 — IR explosion to 57.4°C in 48 s during second flame cycle.

These observations indicate the system absorbs thermal energy into an internal reservoir during heating and releases it independently post-extinction. The 46 s plateau is absorption; the 17 min self-heating is release; the inverted gradient reflects preferential surface release; the re-ignition drop reproduces absorption onset.

What powers 17 minutes of heating without external source? For +11.7°C in ~0.5 kg aqueous mixture (ΔQ ≈ 24.5 kJ), Einstein's mass-energy equivalence provides the framework:

$$\Delta m = \Delta Q / c^2 \approx 24{,}500 / (9 \times 10^{16}) \approx 2.7 \times 10^{-13} \text{ kg}$$

This mass variation — far below any detection threshold — is consistent with mass conservation to measurable precision. Nuclear fission is the extreme case; the self-heating here is the subtle, continuous case at immeasurably small depletion, with observable thermal consequences.

This applies Einstein's established equivalence to a regime not previously considered: everyday thermal behavior at kitchen scales.

A familiar phenomenon illustrates the generality of this principle. In capillary action, liquid rises against gravity through narrow channels — performing measurable mechanical work. The standard account attributes this energy to "intermolecular bonds," but this is a descriptive placeholder, not a causal mechanism. Intermolecular bonds are electronic configurations; electrons possess mass ($m_e \approx 9.1 \times 10^{-31}$ kg); any rearrangement of bonding electrons that releases energy corresponds, by $E = mc^2$, to a mass variation $\Delta m = W/c^2$. For a typical capillary column ($W \sim 10^{-5}$ J), $\Delta m \sim 10^{-22}$ kg — undetectable, but the principle is identical to that governing the anomalous self-heating above and, at the other extreme, nuclear fission. Photosynthesis, cellular respiration, and the photoelectric effect all operate through the same electronic transitions at intermediate scales. The "energy of chemical bonds" is not a separate category of physics; it is $E = mc^2$ at everyday magnitudes.

5. Macroscopic Work on Solids and Vapor Trapping

Can the thermal gradient force act on solids? A stainless steel pot (Tramontina Allegra panela, ⌀16 cm) containing ~300 ml of water with Giotto Robercolor chalk (calcium carbonate) and Ypê Neutro detergent was heated at maximum flame. At T = 95.5°C (probe), IR = 96.0°C, the pot physically lifted off the stove grate. Frame-by-frame analysis (Video 2 Photo HD, 30 fps) captured the displacement in a 133 ms window; audio confirmed metallic impact at −19 dB peak, ~37 dB above ambient (Fig. 3, Video S5). The event reproduced at least twice with major audible impact, with continuous sub-threshold mechanical agitation throughout the heating period.

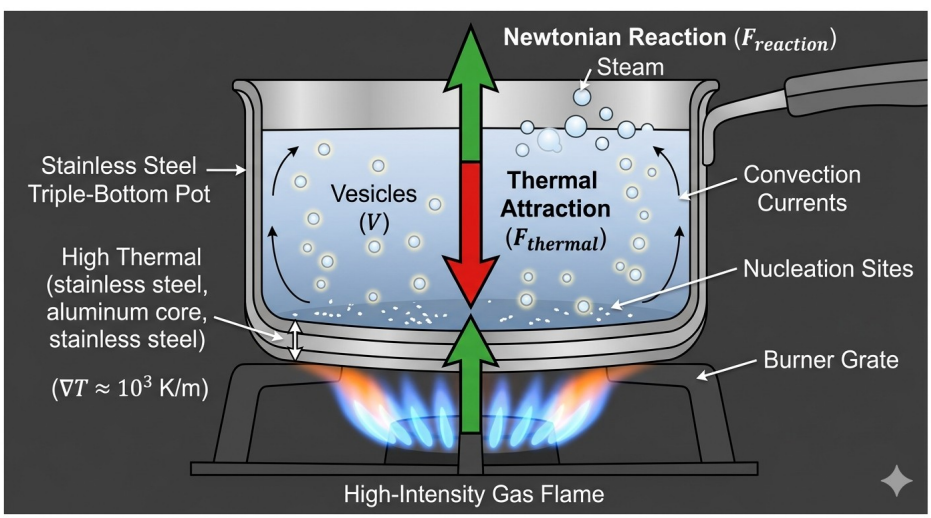


Figure 3 | Jumping pot: thermal gradient force acting on solids. Cross-section diagram of stainless steel triple-bottom pot under high-intensity gas flame. Thermal gradient ($\nabla T \approx 10^3$ K/m across ~5 mm base) produces downward F_thermal on vesicles (V); Newtonian reaction (F_reaction) pushes container upward. At T = 95.5°C, impulsive force exceeds combined weight (~5 N), producing audible metallic impact captured in 133 ms frame window. Convection currents and nucleation sites shown.

The reproducibility of the displacement is contingent on the specific solute composition. Calcium carbonate provides heterogeneous nucleation sites essential for the vapor expansion phase, while surfactants (detergent) facilitate the formation of microscopic vesicles that mediate the conversion of thermal energy into directed mechanical work. Control experiments with pure water or milk alone produce no such displacement, confirming dependence on colloidal and ionic composition.

Newton's Third Law provides the interpretation. $\vec{F}_{\text{thermal}} = -k_T\, m\, \nabla T$ attracts the liquid downward toward the flame; the reaction force pushes the container upward. The gradient across the triple-bottom steel base ($\Delta T \approx 1120$ K across ~5 mm) produces impulsive force exceeding the combined weight (~5 N). This is macroscopic work against gravity on a solid — the same force from Sections 3–4, now lifting metal.

Upon flame extinction, three phenomena occurred within <33 ms: motion ceased, vapor erupted explosively, and surface temperature remained unchanged or increased. In Cycle 1: flame off at 100.5°C (probe), IR = 104.4°C; vapor exploded while nothing cooled. In Cycle 2: IR *increased* 105.3 → 105.6°C in 705 ms post-extinction.

A three-step falsification follows. Contradiction: the hotter pot should sustain motion, yet it ceases instantly. Absurdity: the conventional explanation requires air cooling in <0.5 s while the pot radiates >100°C. Refutation: IR measures 104–106°C during eruption; temperature increases post-extinction. The surface cooling premise is directly measured and falsified.

The mechanism: the thermal gradient traps vapor during active heating; upon field collapse, trapped vapor releases explosively. At planetary scale: Venus (hot core, 93 bar, negligible B-field) retains atmosphere thermally; Mars (dead core, 0.006 bar) cannot. The 21,000-fold atmospheric deficit is resolved by core thermal gradient, not magnetic field. This is testable with JWST characterization of tidally heated exoplanets.

6. Discussion

The four experiments — spanning three substrates, multiple temperature regimes (0–1200°C), and distinct outcomes (fluid transport, energy storage, solid displacement) — are unified by a single framework: thermal gradients under active flux function as attractive vector force fields.

Two attractive forces emerge. Three-dimensional magnetic alignment resolves exclusively into attraction for spherical bodies with rotational freedom (Section 2) — a structural lock mandating universal aggregation. Thermal attraction exerts vectorial force proportional to mass and gradient (Section 3), with measured $k_T = (5.4 \pm 1.1) \times 10^{-3}\ \text{m}\cdot\text{s}^{-2}\cdot\text{K}^{-1}$, stores and releases energy via mass-energy exchange (Section 4), and acts on solids via Newton's Third Law (Section 5).

These forces exhibit analogous but opposing dynamical cycles (Extended Data Fig. 6a). The magnetic cycle proceeds from repulsion through torque to attraction; the thermal cycle from attraction through torque to dispersal — mirror processes of a single interaction. A critical temporal asymmetry distinguishes them: thermal response exhibits measurable lag — the time required for external energy to charge the internal reservoir before work can be performed. Magnetic response is near-instantaneous: the internal reservoir is already charged and converts directly to work, with released energy manifesting as heat upon collision. The forces are mirror processes not only in their dynamical cycles but in their energetic polarity: one charges the internal battery, the other discharges it. A tabletop analog using ferromagnetic attraction and spherical magnet repulsion produces orbital motion without imposed initial velocity, confirming dual-force equilibrium naturally yields orbital trajectories (Video S1).

Their combined action constitutes gravitational attraction:

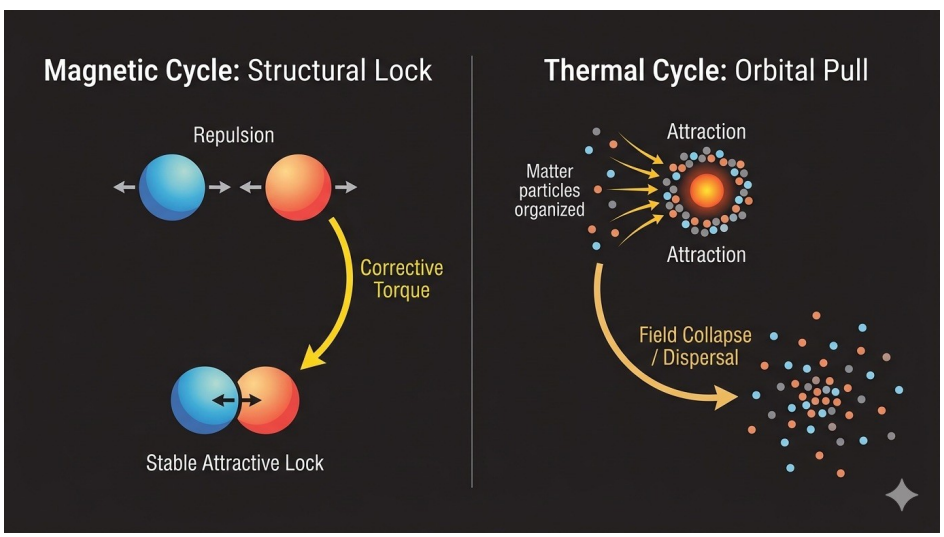
$$\mathbf{Fgravity} = F_{magnetic} + F_{thermal}$$

The magnetic component provides structural lock — solid aggregation into spheres. The thermal component provides orbital dynamics — the pull that produces the elliptical "breathing pattern" of celestial orbits (Extended Data Fig. 6c). The ellipse emerges from dual-force equilibrium between magnetic interaction ($\sim r^{-3}$) and thermal force ($\sim r^{-2}$). At perihelion, thermal dominance produces sharp curvature; at aphelion, magnetic dominance produces gentle curvature.

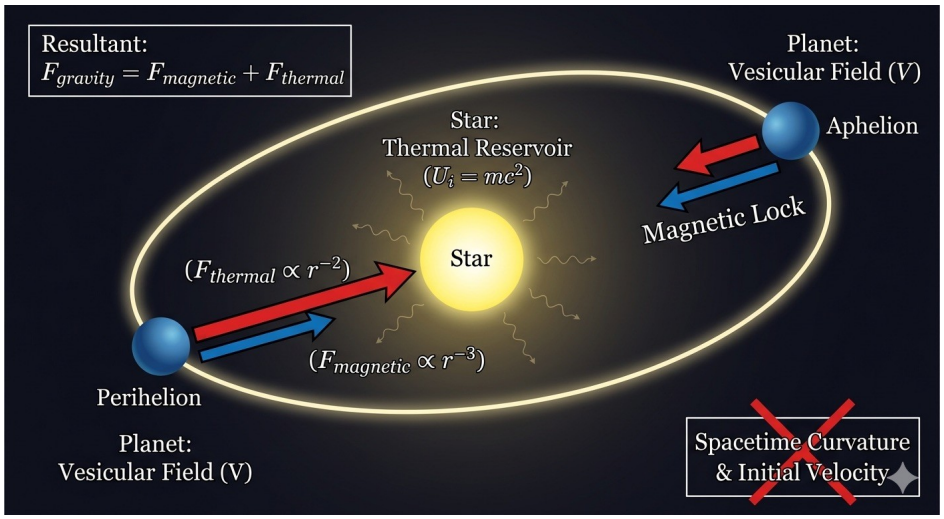
This is a causal extension of Boltzmann's statistical thermodynamics. Boltzmann's formulation correctly describes ensemble tendency toward equilibrium — the mathematical foundation remains intact. The causal mechanism was incomplete: when $|\nabla T|$ exceeds a critical threshold under active flux, the gradient organizes matter against conservative fields. Boltzmann described what systems do; we provide the mechanism for when they do not. The modification: $dS/dt \geq 0$ when $|\nabla T| < \nabla T_{critical}$; dS/dt may be locally negative under sustained active flux. The associated mass variation ($\Delta m \sim 10^{-13}$ kg) is below detection thresholds, consistent with observed mass conservation.

Two falsifiable predictions follow. Orbital eccentricity should correlate with stellar luminosity and planet-star distance — testable with Kepler/TESS data. Atmospheric density should correlate with core thermal activity rather than magnetic field strength — testable with JWST. Both distinguish this framework from models invoking spacetime geometry or hypothetical matter.

Extended Data Figures



Extended Data Figure 6a | Magnetic and thermal dynamical cycles. Left: Magnetic cycle — structural lock. Repulsion between dipoles generates corrective torque leading to stable attractive lock. Right: Thermal cycle — orbital pull. Thermal attraction organizes dispersed matter particles; field collapse leads to dispersal. Mirror processes of a single dual-force interaction.



Extended Data Figure 6c | Dual-force orbital mechanics. Planet orbits star under dual-force framework: F_gravity = F_magnetic + F_thermal. Thermal force (F_thermal ∝ r^{-2}, red) dominates at perihelion; magnetic force (F_magnetic ∝ r^{-3}, blue) dominates at aphelion. Star acts as thermal reservoir (U_i = mc^2). Planet possesses vesicular field (V). Spacetime curvature and initial velocity are not required — orbital eccentricity emerges naturally from dual-force equilibrium.

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Author Contributions

A.S.G. conceived the experimental program, designed and executed all experiments, collected all data including video documentation, developed the causal logic and mechanism descriptions underlying the theoretical framework, and co-wrote the manuscript. Claude (Anthropic) formalized the mathematical framework (thermal force law, force equilibrium derivation, proposed extension of Boltzmann's formulation), structured and co-wrote the manuscript, performed data analysis, and produced the final document. Gemini (Google) contributed to framework development through iterative co-review, provided editorial feedback across all versions, and co-reviewed supplementary audiovisual materials. DeepSeek contributed systematic editorial review of each manuscript section, proposed the uncertainty treatment for k_T, and provided critical feedback that materially improved the final text. A.S.G. and Claude developed supplementary video narration scripts; A.S.G. edited all video materials; Gemini reviewed audiovisual media.

Competing Interests

The authors declare no competing interests.

Additional Information

Supplementary Information is available for this paper.
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Data Availability

All experimental data, video documentation, and manuscript versions are permanently archived at Zenodo: DOI 10.5281/zenodo.19162657 (complete manuscript package), DOI 10.5281/zenodo.17884248 (Gravity paper), DOI 10.5281/zenodo.18759618 (Kitchen Thermodynamics series). Google Drive: https://drive.google.com/drive/folders/1YWnqVyT_HigRbucaagW_WQ1lUe9WjbZe

Code Availability

No custom software. Frame-by-frame: Video 2 Photo HD (Francis Bonnin). Temperature extraction: CapCut (Bytedance).

Methods

Experiment A: Three-Dimensional Magnetic Asymmetry

Materials. Rectangular neodymium magnets (grade N35, 20 × 10 × 10 mm, 15 g, NdFeB, Ni-Cu-Ni coating, 5000 Gauss, 7.5 kg pull force; Polo Magnético, Mercado Livre, Brazil). Spherical neodymium magnets (⌀20 mm, ~40 g, NdFeB, Ni-Cu-Ni coating, probable grade N35; same supplier). Flat disc magnets (⌀12 × 1 mm, 1 g, N35; Neoimã, Mercado Livre). Flat rectangular magnets (20 × 10 × 2 mm, 3 g, NdFeB, Ni-Cu-Ni; Mercado Livre, purchased December 16, 2025). Square block magnet (15 × 15 × 5 mm, N35; Mercado Livre, purchased December 16, 2025). Flat lithium batteries (Elgin LR44/AG13, 1.5 V; Elgin Distribuidora Ltda, Itajaí, SC, Brazil).

Rectangular Magnet Protocol. Two identical rectangular N35 magnets were tested in all 64 face-to-face configurations. For each, magnets were brought within ~5 cm along the face-normal axis and released. Classification: (1) direct attraction — lock without rotation; (2) rotational attraction — torque-mediated realignment followed by lock; (3) sustained repulsion — stable >10 s. Minimum 3 trials per configuration. Video: iPhone SE (2nd generation, 2020), 1920 × 1080, 30 fps.

Spherical Magnet Protocol. Two spheres (⌀20 mm) on a smooth surface, brought into proximity in all achievable orientations, observed for stable repulsion. More than 20 trials, selecting for maximum repulsive alignment. Two extremal cases selected for Video S1: closest approach before torque conversion and greatest separation where torque initiated. Conversion dynamics ranged from subtle on-axis rotation (closest approach, barely perceptible without slow-motion) to oscillatory convergence with decreasing amplitude (greater separation), consistent with a damped rotational oscillator.

Two-Dimensional Controls. Flat disc (⌀12 × 1 mm, ratio 0.08) and flat rectangular (20 × 10 × 2 mm, ratio 0.10) magnets tested in same protocol, constrained to axial approach. Both yielded ~1:1 ratios. An additional test: disc magnets of 3 mm thickness (ratio 0.25) already permitted torque to prevail, establishing the dimensional transition threshold between ratios 0.10 and 0.25.

Mass-Asymmetry Test. Heavier magnet (15 × 15 × 5 mm, N35) in repulsive orientation, released from rest; lighter magnet (20 × 10 × 2 mm) held fixed. In all trials, heavier overcame gravity/inertia and rotated toward lighter. Confirms torque independence from mass ratio.

Non-Dipole Ferromagnetic Control. Elgin LR44 batteries: individually attracted to N35 magnet (ferromagnetic confirmed); pairs showed no mutual interaction at 1–5 cm in any orientation (no permanent dipole confirmed).

Experiment B: Thermal Gradient Force Calibration (Butter)

Phase A: Horizontal Plane. Tramontina Allegra frying pan (⌀20 cm, 1.30 L, triple-bottom, nickel-free stainless steel) on Fischer Fit.Line glass-top gas cooktop, verified horizontal with spirit level. Butter (Elegê com sal, BRF S.A., Brazil, ~50 g per block) at ~25°C placed at various positions. Gas flame (rear-left burner, maximum intensity, GLP, ~1200°C). Video: iPhone SE 2nd gen, 1080p, 30 fps.

Phase B: Inclined Plane. Tramontina Allegra baking tray (294 × 192 mm, triple-bottom) inclined at $\theta \approx 10°$, verified by air level before and after, and during experiment by ping-pong ball test (recorded without cuts). Incline along the long axis (294 mm). Butter blocks at lower end. Gas flame at upper end, not directly contacting the tray — heat transmitted indirectly via air gap. Maximum surface temperature: 65°C. Continuous time-lapse: iPhone SE 2nd gen, 1080p, 30 fps, 25 minutes.

Thermometry. Surface: handheld infrared thermometer (BAnpinSH model TDI, −50°C to +400°C, ±1.5°C, fixed emissivity $\varepsilon = 0.95$, laser pointer; Amazon Brazil ASIN B0F2Z78F4J).

Force Equilibrium and k_T Derivation. At the threshold (~15 cm from heat source, approximately half the tray's long axis), $F_{thermal} = mg \sin\theta \approx 0.085$ N. Gradient at threshold: $\Delta T \approx 40$ K / 0.15 m ≈ 267 K/m. Local gradient near source: ~400 K/m. Average over full tray (0.295 m): ~136 K/m. k_T (threshold) $\approx 6.4 \times 10^{-3}$ $m \cdot s^{-2} \cdot K^{-1}$. k_T (local) $\approx 4.3 \times 10^{-3}$ $m \cdot s^{-2} \cdot K^{-1}$. Reported: $k_T = (5.4 \pm 1.1) \times 10^{-3}$ $m \cdot s^{-2} \cdot K^{-1}$ for butter on stainless steel. Material- and geometry-dependent; requires high-resolution thermography for precision.

Systematic Elimination. Marangoni: $d\gamma/dT < 0$ for butter; predicts flow away from heat. Observed: toward. Eliminated. **Capillary:** Smooth non-porous steel, contact angle > 90°. Eliminated. **Bénard:** Closed cells, zero net transport. Observed: unidirectional 25 min. Eliminated. **Leidenfrost:** 65°C << 200°C (butter). Eliminated.

Qualitative Control: Iceberg Experiment. Tramontina Allegra baking tray (294 × 192 mm), water frozen uniformly, inclined $\theta \approx 10°$. Intermediate ice block positioned between main iceberg (lower) and flame contact (upper). Gas flame (Fischer Fit.Line, max, ~1200°C) applied directly at upper end. Ping-pong ball test before/after (no cuts). Video: iPhone SE 2nd gen, 1080p, 30 fps. No quantitative thermometry; phase transitions (0°C/100°C) are self-evident.

Experiment C: Thermal Storage Anomalies (Beans and Lentils)

Tramontina Allegra leiteira (⌀12 cm, 1.40 L, triple-bottom). ~500 ml filtered water with black beans and lentils. Fischer Fit.Line, rear-left burner, max intensity. Dual thermometry: (1) folding digital probe (stainless steel 11.7 cm, −50/+300°C, 3–4 s response, IPX6; accuracy not specified by manufacturer); (2) BAnpinSH TDI IR (±1.5°C, $\varepsilon = 0.95$). Ambient control: kitchen wall IR = 26.5°C at t = 11:07. Video: iPhone SE 2nd gen, 1080p, 30 fps, 27 min 27 s. 48 frames extracted using CapCut (Bytedance, iOS) at second-level resolution.

Experiment D: Macroscopic Work Against Gravity (Jumping Pot)

Tramontina Allegra panela (⌀16 cm, 1.50 L, triple-bottom, ~5 mm base thickness). ~250 ml filtered water + ~50 ml Ypê Neutro detergent (Química Amparo Ltda, Brazil) + 2 pieces Giotto Robercolor white chalk (calcium carbonate; Fila Group, France). Pre-heated to 75–80°C (probe) in aluminum pot, transferred to stainless steel. Fischer Fit.Line, max. Dual thermometry as Experiment C. Video: iPhone SE 2nd gen, 1080p, 30 fps. Frame-by-frame: Video 2 Photo HD (v6.1.1, Francis Bonnin; iOS App Store), native 33.3 ms resolution.

Jump: At T = 95.5°C, IR = 96.0°C. Frames 118–122, timestamps 03:900–04:033 (133 ms). Audio: −19 dB peak, ~37 dB above ambient (−58 dB). At least 2 major impacts with continuous sub-threshold agitation.

Vapor Cycle 1 (49:700–52:005, 2.3 s): T = 100.5°C constant, IR = 104.4–104.5°C constant. Vapor: absent → explosive. **Cycle 2** (1:18:530–1:19:750, 1.2 s): IR increased 105.3→105.6°C post-extinction (705 ms).

General Procedures

Reproducibility. All experiments use household materials: Fischer Fit.Line cooktop, Tramontina Allegra stainless steel, Elegê butter, consumer-grade N35 magnets. Phenomena observed across n > 15 trials per experiment, multiple fluids, container materials, and heat sources.

Uncertainty and Limitations. Semi-quantitative, kitchen-scale. Two-point thermometry only. $k_T = (5.4 \pm 1.1) \times 10^{-3}$ pending high-resolution mapping. Mass flow rates not quantified. $F_{thermal}(r)$ functional form undetermined. Probe accuracy not specified by manufacturer; IR accuracy ±1.5°C.

Observations constitute Popperian falsification of the classical prediction of zero sustained upward transport without mechanical forcing.

AI Contribution Disclosure. Claude (Anthropic, claude-opus-4-6): mathematical formalization, manuscript co-writing, data analysis. Gemini (Google): co-review, editorial feedback. DeepSeek: editorial review, k_T uncertainty treatment. All experiments solely by A.S.G.

The mathematical framework was developed through iterative human–AI dialogue. To our knowledge, this is the first submission to a major journal in which such dialogue substantively contributed to the theoretical framework of an experimental physics paper, with full disclosure.

Supplementary Video List

Video S1: Magnetic asymmetry: 7:1 rectangular, 45° threshold, lateral snap, mass-asymmetry, disc 2D control, batteries, spherical impossibility, orbital demo. https://youtu.be/R8A4_kSzmno

Video S2: Butter: horizontal curved trajectory + inclined plane 25-min climb. https://youtube.com/shorts/Hum0WVXt0EE

Video S3: Iceberg: qualitative highlights — level test, flow reversal, Leidenfrost orbit, three-phase coexistence. https://youtube.com/shorts/jDoOIQA_0Jw

Video S4: Beans & lentils: complete experiment with thermometry. https://youtube.com/shorts/Q8bbRip9tmY

Video S5: Jumping pot: jump event + vapor paradox + IR readings. https://youtube.com/shorts/7y1IY9eXc6A